A CASE-STUDY BASED REPORT

ON

Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

Submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

A.Y.: 2026–27

Submitted by

CH.V.S. Manikanta Varma Raju Roll No.: 2520030239

Ishaan Kulkarni Roll No.: 2520030267

Team Number: 18

Under the Esteemed Guidance of

Mr. Shaik Asif

Assistant Professor, Department of CSE

Koneru Lakshmaiah Education Foundation (KLH), Hyderabad

INDEX

Abstract

1. Introduction

   1.1 Background

   1.2 Motivation for the Project

   1.3 Overview of the Work Carried Out

   1.4 Scope and Responsible-Use Statement

2. Problem Statement

   2.1 Case-Study Scenario

   2.2 Central Case-Study Question

   2.3 Formal Problem Definition

   2.4 Objectives that Guide the Study

   2.5 Assumptions and Limitations

3. Dataset and Exploratory Data Analysis

   3.1 About the Dataset

   3.2 Purpose of EDA

   3.3 Preparation Steps Actually Performed

   3.4 EDA Exploration – Implementation

   3.5 Summary of EDA Findings

4. Algorithm Selection and Implementation

   4.1 Why Logistic Regression

   4.2 Why Elastic Net Regularization

   4.3 Preprocessing and Data Split

   4.4 Training and Hyperparameter Tuning

   4.5 Evaluation Metrics and Results

   4.6 Website Demonstration

5. Conclusion

References

Team Member Signatures

Abstract

Customer churn occurs when a customer stops using a company's service. Identifying customers who may leave can help a business decide where retention efforts may be useful. This case study applies Elastic Net-regularized Logistic Regression to the Telco Customer Churn dataset. The dataset contains 7,043 customer records and a binary Churn label. The workflow includes exploratory data analysis, cleaning the TotalCharges field, imputing missing values, encoding categorical variables, standardizing numerical variables, a stratified train/test split, and five-fold cross-validation to tune model parameters. On the held-out test set, the recorded model achieved 80.06% accuracy, 64.67% precision, 54.81% recall, 59.33% F1-score, and 84.07% ROC-AUC. A local website interface was developed to demonstrate customer-record exploration and model predictions. The results show that the model can rank churn risk usefully, but its recall also indicates that some customers who churn are missed. The project is an educational prototype, not a production retention system.

1. Introduction

1.1 Background

Customer churn is a relevant business problem for subscription-based services such as telecommunications. When customers discontinue a service, the provider may lose recurring revenue and may need to spend resources acquiring replacement customers. Machine learning can use historical customer information to estimate which customers are more likely to churn.

1.2 Motivation for the Project

The project explores whether demographic, service, contract, and billing information can support churn-risk prediction. The chosen approach aims to balance predictive performance with interpretability: Logistic Regression provides a probability-based binary classifier, while Elastic Net regularization constrains coefficients and can reduce the influence of weak or redundant features.

1.3 Overview of the Work Carried Out

- Reviewed the Telco Customer Churn dataset and examined its shape, columns, data types, missing values, class distribution, and selected feature relationships.
- Converted TotalCharges to numeric values and handled its 11 blank entries through median imputation in the numerical preprocessing pipeline.
- Excluded customerID from model inputs because it is an identifier rather than a meaningful customer characteristic.
- Built a scikit-learn pipeline for numerical imputation and standardization, categorical imputation and one-hot encoding, and Elastic Net Logistic Regression.
- Used a stratified 80/20 train/test split and GridSearchCV with five-fold cross-validation to tune C and l1_ratio using ROC-AUC.
- Evaluated the selected model on the held-out test set and built a Flask-based local website to demonstrate the saved model and customer churn prediction.

1.4 Scope and Responsible-Use Statement

This project demonstrates a supervised binary-classification workflow using one public-style telecom churn dataset. Predictions are estimates based on patterns in the available data; they do not prove that a customer will leave or explain the cause of an individual customer's behaviour. The model should not be used as the sole basis for customer decisions without further validation, monitoring, privacy review, and evaluation on current business data.

2. Problem Statement

2.1 Case-Study Scenario

A telecommunications provider has historical customer records containing contract details, service subscriptions, tenure, and charges. The provider wants a data-driven way to identify customers with a higher estimated probability of churn so that staff can consider appropriate retention actions.

2.2 Central Case-Study Question

Can an Elastic Net-regularized Logistic Regression model use the available customer attributes to distinguish customers labelled as churners from customers labelled as non-churners on unseen test data?

2.3 Formal Problem Definition

Given customer feature vectors X containing demographic, service, contract, and billing attributes, predict the binary target y = Churn, where No is encoded as 0 and Yes as 1. The model estimates $P(y = 1 \mid X)$, the probability associated with the churn class. A classification threshold converts that probability into a predicted class.

2.4 Objectives that Guide the Study

- Understand the dataset and identify data-quality issues before modelling.
- Build a repeatable preprocessing and modelling pipeline.
- Apply Elastic Net regularization to combine L1 and L2 penalties.
- Tune model hyperparameters using cross-validation rather than selecting them solely from test performance.
- Report several classification metrics, with particular attention to recall and ROC-AUC because the target classes are imbalanced.
- Demonstrate the saved model through a local website interface.

2.5 Assumptions and Limitations

- The dataset's historical labels are treated as the target for supervised learning.
- The recorded test metrics apply to the particular split and model run used in the notebook; they are not a guarantee of performance on future customers.
- The dataset has more non-churners than churners, so accuracy alone would not describe performance adequately.
- A fixed dataset may not reflect changes in customer behaviour, pricing, or service offerings over time.
- Feature associations are predictive relationships, not proof of causation.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project uses the Telco Customer Churn CSV file. It contains 7,043 rows and 21 columns, including customerID, demographic attributes, subscribed services, contract and payment details, tenure, MonthlyCharges, TotalCharges, and the Churn target. The target distribution is 5,174 No records (about 73.46%) and 1,869 Yes records (about 26.54%).

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) was used to understand the structure and quality of the data before training. It helps reveal missing or incorrectly typed values, class imbalance, distributions, and relationships that affect preprocessing and evaluation choices. EDA is an analysis stage within the machine-learning lifecycle, not a separate prediction algorithm.

3.3 Preparation Steps Actually Performed

- Inspected the first records, dataset dimensions, column names, data types, and descriptive statistics.
- Converted TotalCharges from text to numeric values; 11 blank entries became missing values and were handled by median imputation in the pipeline.
- Checked duplicate records and customer identifiers; the dataset had no duplicate rows and no duplicate customer IDs in the EDA checks.
- Removed customerID from the predictors to avoid treating an identifier as a behavioural feature.
- Encoded the target as 0 for No and 1 for Yes.

3.4 EDA Exploration – Implementation

The analysis examined the churn class distribution, numerical summaries and distributions, tenure and monthly charges by churn label, churn rates by contract and internet-service category, and correlations among tenure, MonthlyCharges, and TotalCharges. These checks informed the handling of missing values, categorical encoding, numerical standardization, and the use of multiple evaluation metrics.

3.5 Summary of EDA Findings

- The churn class is imbalanced: approximately 26.54% of records are labelled Yes.
- Customers labelled as churners had lower average tenure than non-churners in this dataset (about 17.98 months versus 37.57 months).
- Average MonthlyCharges were higher among churners (about 74.44) than non-churners (about 61.27), in the dataset's recorded units.
- Historical churn rates differed across contract categories: month-to-month was about 42.7%, one-year about 11.3%, and two-year about 2.8%.
- Historical churn rates also differed by internet-service category: fiber optic about 41.9%, DSL about 19.0%, and no internet service about 7.4%. These are associations and do not establish causation.
- Tenure and TotalCharges were strongly positively correlated (about 0.826), which is one reason coefficient regularization was worth investigating.

4. Algorithm Selection and Implementation

4.1 Why Logistic Regression

The target has two classes, Churn = Yes or No, so this is a supervised binary-classification problem. Logistic Regression estimates a probability between 0 and 1 using the sigmoid function, then assigns a class using a threshold. Despite the word "regression" in its name, it is commonly used for classification. Its coefficients also make the direction of feature associations easier to inspect than in many more complex models.

4.2 Why Elastic Net Regularization

Elastic Net combines L1 and L2 regularization. The L1 component can shrink some coefficients to zero, providing a form of feature selection. The L2 component shrinks coefficients and can improve stability when

predictors are correlated. The combined penalty is useful to test on customer data with multiple related service and billing attributes. This choice does not guarantee that Elastic Net will outperform every alternative.

The objective can be written as:

$$\text{Log-loss} + \lambda \left[\alpha \Sigma|\beta_j| + (1 - \alpha) / 2 \cdot \Sigma\beta_j^2 \right]$$

Here, λ controls the overall regularization strength and α controls the balance between the L1 and L2 penalties. In scikit-learn's LogisticRegression implementation, l1_ratio controls the mix and C is the inverse of regularization strength. The fitted model used solver='saga', penalty='elasticnet', and a search over C and l1_ratio.

4.3 Preprocessing and Data Split

- The dataset was split into training and testing subsets in an 80/20 ratio using stratification to preserve the target-class proportions.
- Numerical features used median imputation followed by StandardScaler.
- Categorical features used most-frequent imputation followed by OneHotEncoder with unknown categories ignored.
- Preprocessing was combined with the classifier in a scikit-learn Pipeline. During cross-validation, the preprocessing steps are fitted on each training fold rather than using the validation fold to fit transformations, helping reduce data leakage.
- The target column was kept separate from the input features; customerID was excluded.

4.4 Training and Hyperparameter Tuning

GridSearchCV evaluated combinations of C = 0.01, 0.1, 1, 10, 100 and l1_ratio = 0, 0.25, 0.5, 0.75, 1.0 using five-fold cross-validation. This corresponds to 25 parameter combinations and 125 fold fits. ROC-AUC was used as the cross-validation scoring metric. The notebook record for the selected configuration states C = 10 and l1_ratio = 0.5. The final estimator was then evaluated on the held-out test set.

4.5 Evaluation Metrics and Results

The recorded held-out test results for Elastic Net Logistic Regression were as follows. Values are shown as recorded in the model run, rounded to two decimal places.

Metric	Test result
Accuracy	80.06%
Precision (churn class)	64.67%
Recall (churn class)	54.81%
F1-score (churn class)	59.33%
ROC-AUC	84.07%

Accuracy is the fraction of all test predictions that were correct. Precision measures how many customers predicted as churners were actually labelled churners. Recall measures how many actual churners the model identified. F1-score balances precision and recall. ROC-AUC measures how well the model ranks positive cases above negative cases across classification thresholds. The ROC-AUC result indicates useful ranking ability, while the recall result shows that the model still misses a substantial proportion of actual churn cases at the selected threshold.

The recorded confusion matrix was TN = 923, FP = 112, FN = 169, and TP = 205. In this matrix, TP means a churner correctly predicted as churn, TN means a non-churner correctly predicted as non-churn, FP means a non-churner incorrectly flagged as churn, and FN means a churner missed by the model.

4.6 Website Demonstration

A local Flask-based website was developed as a demonstration layer around the saved model. It includes a sign-in interface, dashboard, customer-record browsing, prediction workflow, and reports/analytics views. The prediction workflow selects a customer record, sends its input features to the backend, and displays the model's predicted churn probability and class. The historical Churn label is not used as an input feature for prediction. The interface is a project demonstration; it does not by itself establish production readiness or real-time integration with a telecom provider.

5. Conclusion

This case study built an end-to-end supervised machine-learning workflow for customer churn prediction using Elastic Net-regularized Logistic Regression. EDA identified class imbalance, a small number of blank TotalCharges values, differences in churn rates across contract and service categories, and correlated numerical features. The pipeline combined imputation, standardization, one-hot encoding, and regularized Logistic Regression, while GridSearchCV selected hyperparameters using five-fold cross-validation. On the recorded held-out test set, the model achieved 80.06% accuracy and 0.8407 ROC-AUC, with 54.81% recall for the churn class. The result is promising as an educational prototype but leaves room to improve the detection of churners. Future work could examine threshold selection based on business costs, class-weighting or resampling strategies, repeated or nested validation, calibration, and testing on newer data. Any claimed improvement should be supported by a new evaluation rather than assumed.

References

- IBM / Kaggle-style Telco Customer Churn dataset, commonly distributed as Telco-Customer-Churn.csv. Dataset fields and counts reported here are from the project's local CSV.
- scikit-learn documentation, LogisticRegression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- scikit-learn documentation, Pipeline and ColumnTransformer: https://scikit-learn.org/stable/modules/compose.html
- scikit-learn documentation, GridSearchCV: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html
- Research paper supplied with the project: Arivazhagan and Sankara Subramanian (2020), Logistic Regression with L1 regularization for churn prediction. The paper is discussed as related work; its results are not presented as results of this project.
- Research paper supplied with the project: s10614-022-10275-1.pdf, concerning mixed L1/L2 regularization in churn-related modelling. Consult the source PDF for complete bibliographic details.

Team Member Signatures

Team Number: 18

1. CH.V.S. Manikanta Varma Raju — Roll No. 2520030239

Signature: _____

2. Ishaan Kulkarni — Roll No. 2520030267

Signature: _____

Guide: Mr. Shaik Asif, Assistant Professor, Department of CSE